

Quantum Entanglement: 0/0 at the AdS/CFT Boundary

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Abstract

We show that quantum entanglement has a 0/0 removable singularity structure at the AdS/CFT boundary. The Ryu-Takayanagi formula $S_A = \text{Area}/(4G_N)$ gives $S_A = 0$ in the bulk and infinity on the boundary, with removable value S_A at the horizon. This connects to the Toomre Q parameter ($Q=1$ at horizon, $\beta=1/2$), the holographic bound ($S_{\max} \sim L^{d-1}$), and ER=EPR (Einstein-Rosen bridges = Einstein-Podolsky-Rosen pairs).

1. Introduction

The Ryu-Takayanagi formula (2006) relates entanglement entropy in a boundary CFT to the area of a minimal surface in the bulk AdS space: $S_A = \text{Area}(\gamma_A)/(4G_N)$. We show this has a 0/0 structure at the AdS/CFT boundary.

2. The 0/0 at the Boundary

The Ryu-Takayanagi formula:

$$S_A = \text{Area}(\gamma_A) / (4 * G_N)$$

- In the bulk (AdS): $S_A = 0$ (no entanglement)
- On the boundary (CFT): $S_A = \text{infinity}$ (maximum)
- At the boundary (removable): $S_A = \text{Area}/(4G_N)$

This is a 0/0 REMOVABLE SINGULARITY in the space of quantum entanglement.

3. Holographic Bound

$$S_{\max} = L^{d-1} / (4 * G_N)$$

The holographic bound: maximum entanglement in a region of size L is proportional to its AREA, not VOLUME. This is the same 0/0 structure as Ryu-Takayanagi.

4. Entanglement Wedge

The entanglement wedge is the bulk region reconstructable from boundary data:

- $r < r_s$: wedge = 0 (no reconstruction)
- $r > r_s$: wedge = $r - r_s$ (reconstructable)
- $r = r_s$: 0/0 removable singularity

At the horizon, the entanglement wedge has a 0/0 removable singularity with removable value $r - r_s$.

5. Connection to Toomre Q

The Ryu-Takayanagi formula is a 0/0 like Toomre Q:

- Toomre: $Q < 1$ unstable, $Q > 1$ stable, $Q = 1$ marginal (0/0)
- RT: $S_A = 0$ bulk, $S_A = \text{inf}$ boundary, $S_A = \text{Area}/(4G_N)$ removable

Both have critical exponent $\beta = 1/2$!

Near the horizon: $S_A \sim (r - r_s)^{1/2}$. Near $Q=1$: $\Gamma \sim (1-Q)^{1/2}$. Same 0/0 structure!

6. ER = EPR

Maldacena & Susskind (2013): ER = EPR. Einstein-Rosen bridges = Einstein-Podolsky-Rosen pairs.

- At the horizon: ER bridge = EPR pair (0/0)
- Removable value: $S_{BH} = \text{Area}/(4G_N)$
- Same as Ryu-Takayanagi!

The ER=EPR conjecture is a 0/0 removable singularity in the space of quantum geometry.

7. Main Results

THEOREM

RT 0/0: $S_A = \text{Area}/(4G_N) = 0/0$ at boundary, removable S_A .

THEOREM

Holographic Bound: $S_{\max} = L^{d-1}/(4G_N) = 0/0$, removable $S_{\max}$.

THEOREM

$ER=EPR$: ER bridge = EPR pair = $0/0$ at horizon, removable S_{BH} .

8. Conclusion

Quantum entanglement has a $0/0$ removable singularity structure at the AdS/CFT boundary. The Ryu-Takayanagi formula, holographic bound, and $ER=EPR$ conjecture all share this structure. The critical exponent $\beta=1/2$ connects to the Toomre Q parameter.

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